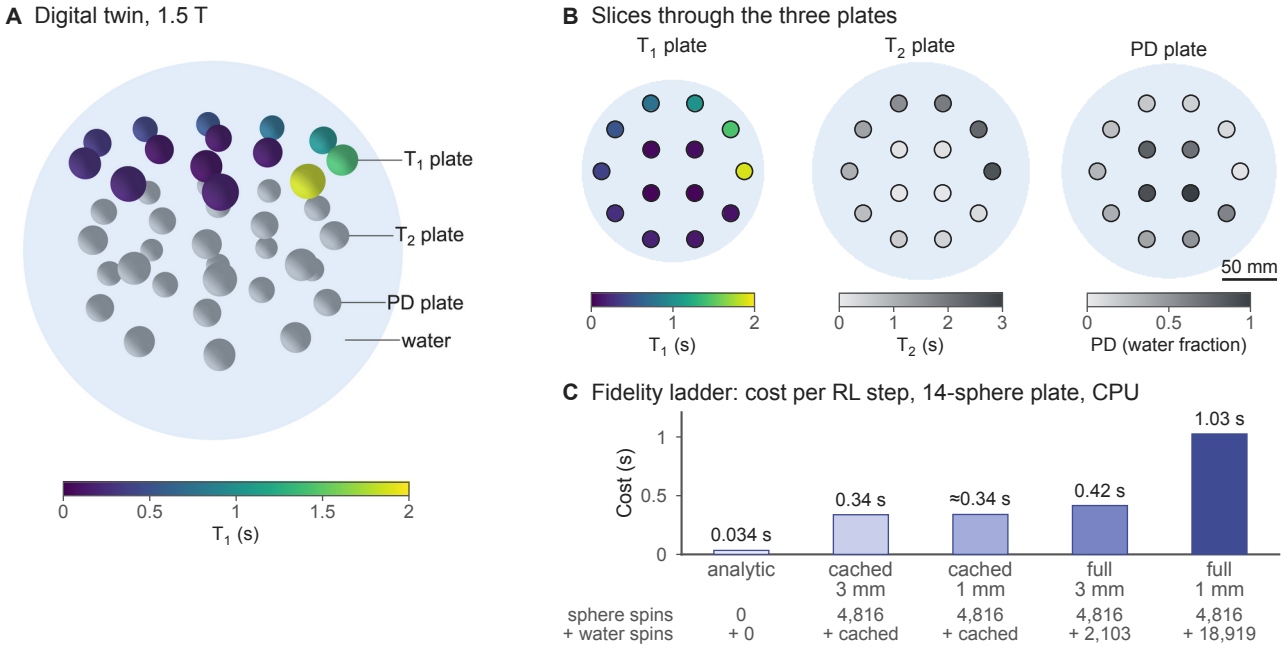

A Digital twin, 1.5 T
T1 plate
T2 plate
PD plate
water
0
0.5
1
1.5
2
T1 (s)
B Slices through the three plates
T1 plate
T2 plate
PD plate
50 mm
0
1
2
T1 (s)
0
1
2
3
T2 (s)
0
0.5
1
PD (water fraction)
C Fidelity ladder: cost per RL step, 14-sphere plate, CPU
Cost (s)
0
0.5
1
0.034 s
0.34 s
≈0.34 s
0.42 s
1.03 s
analytic
cached 3 mm
cached 1 mm
full 3 mm
full 1 mm
sphere spins + water spins
0 + 0
4,816 + cached
4,816 + cached
4,816 + 2,103
4,816 + 18,919